

CYBERSPACE PIRATES

Outsourcing Cyberwar in the Age of AI
and Ransomware

HACKING THE HACKERS



SAT: MAXAR-9
MODE: EO/IR
RES: 0.31m

STARLINK TERMINAL
UPLINK ACTIVE

CRYPTO WALLETS

bc1q7f3u8m2x0w3y8z9v5d6kzq2r3t8l
bc1q4n6x9h2p7m5w0v8y3z2j6d9s4k1
0x9fA3bC5dE7a8B2cD4eF6a87cB09eE0F
0x7cB4d9E6fA2bC8d1E5f7A9c3B6d8E1f
bc1q0l9w5n2y7x3az8v6u4t1s9p2q8r6h

ESTIMATED FUNDS

BTC: 1,246.32
ETH: 3,821.17
XMR: 542.71
USDT: 2,183,672.44
TOTAL USD: \$15,780,392

COMMAND & CONTROL NODES



ACTIVE SESSIONS

185.220.101.8:443	ESTABLISHED
185.220.101.8:8443	ESTABLISHED
91.198.174.6:9001	ESTABLISHED
178.62.10.3:443	ESTABLISHED
64.15.90.57:443	ESTABLISHED

TOOLS

netehac
crackmapexec
nxc
bloodhound
mimikatz

LIVE NETWORK TRACE

23.56.12.9	ESTABLISHED
209.141.35.7	ESTABLISHED
45.77.238.11	ESTABLISHED
185.220.101.8	ESTABLISHED
103.27.6.2	ESTABLISHED
178.62.10.3	ESTABLISHED
64.15.90.57	ESTABLISHED
91.198.174.6	ESTABLISHED

PACKET CAPTURE

0000	45 00 00 00 02 02 00 C10 DC5 B1 F300 B10 000 00
0000	45 00 00 00 55 55 00 B10 000 51 6093 800 000 38
0000	45 00 00 00 02 02 00 C10 DC5 B1 F300 B10 000 00
0000	45 00 00 00 55 55 00 B10 000 51 6093 800 000 38
0000	00 00 00 00 02 02 00 C10 DC5 B1 F300 B10 000 00
0000	00 00 00 00 55 55 00 B10 000 51 6093 800 000 38

WHY I'M TALKING ABOUT PIRATES



PENUMBRA

- Founder & CEO, Penumbra Strategies | CISSP
- 3× White House (2× NSC, OMB)
- Executive Orders on emerging tech, cyber, and digital assets
- Regulatory advisor (CFTC, New York, Idaho) + California Governor's Innovation Council



1LT Carole House
Regulator 6
2nd PLT, 45th MICO, 2-1 CAV
March 2013 - April 2014



WHY THE KRAKEN IS WINNING

The Scale Mismatch (\$, people, time)

The Operational Reality (reach and positioning)

The Sanctuary Wall (extradition challenges)

**BLUF: an architectural mismatch
between democratic law and
adversarial scale undercuts
deterrence**



DEAD RECKONING: THREE SCHISMS, ONE SHIP

This is a live, unresolved policy question, with disagreement on many sides. Illustrative examples:

Sector	FOR: Expand/Delegate	AGAINST: Contain/Government Only
Executive Branch	Trump Admin Cyber Strategy for America (March 6, 2026): “ unleash the private sector by creating incentives to identify and disrupt adversary networks and scale our national capabilities”	White House ONCD adviser Thomas Lind, Prague (March 19, 2026): “That does not mean hack back . That does not mean letters of marque.” (Recorded Future News)
Congress	H.R. 9697/S. 5000 (Burchett/Lee, July 2026): commission private operators for cyberspace ops, up to 15% asset bounty	No federal majority to repeal CFAA. DOJ has consistently enforced 18 U.S.C. §1030 against unauthorized access regardless of target's criminal status.
Industry & Experts	Stewart Baker (former NSA General Counsel): argues CFAA does not clearly prohibit all active defense ; advocates CFAA reform. [**Not support, but a call for rethinking – Matt Pearl (CSIS, 2025): argues capacity gap is real and hack-back authorization deserves reconsideration].	Alexander Klimburg (CSIS, 2025): “ Shooting without aiming – hack back is not a magic bullet.” Devin Lynch (Aspen Digital, May 2026): letters of marque is “ historical cosplay .” Robert Knake (CFR, former NSC): “Instead of hacking back, U.S. companies should let Cyber Command do it for them ”

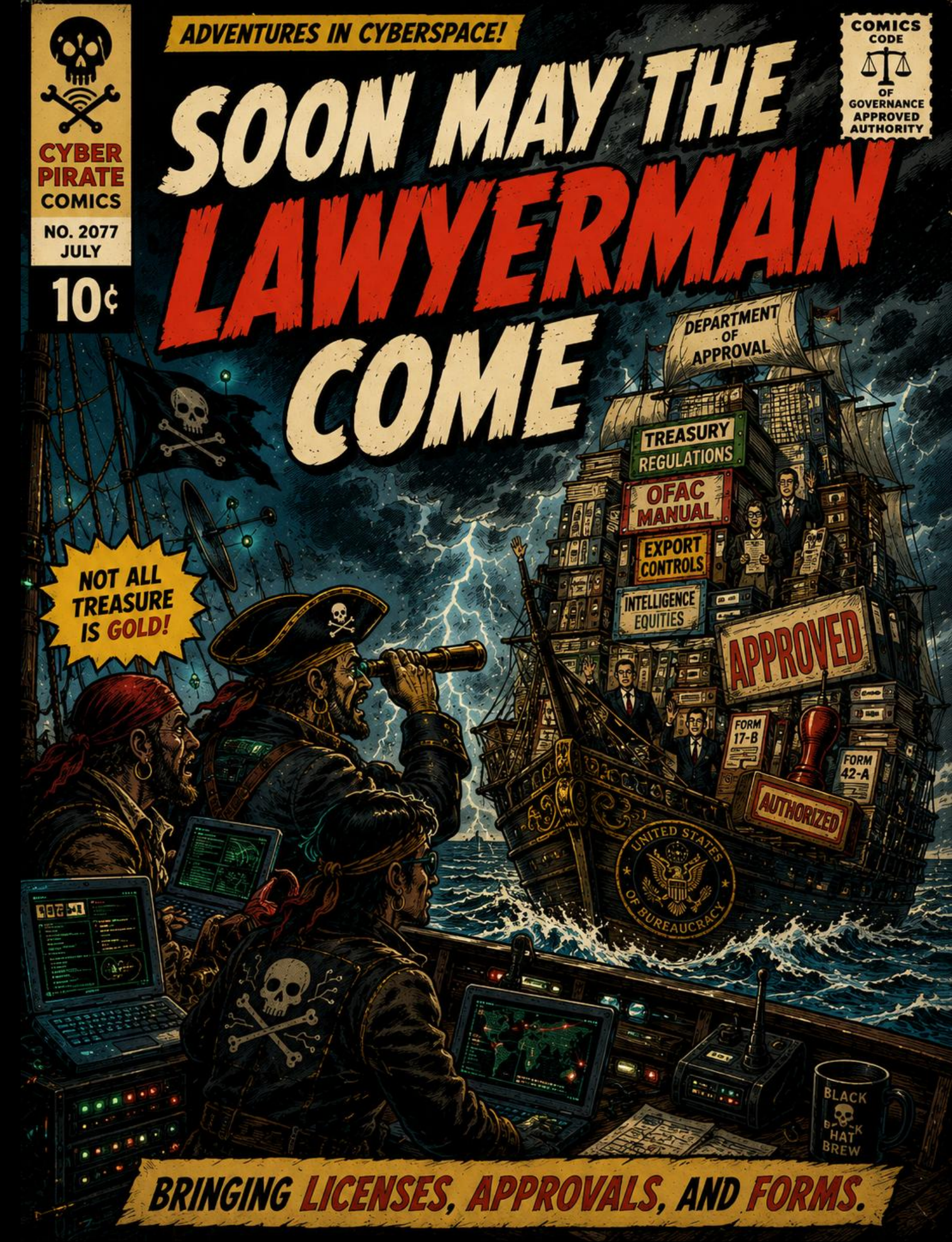
STRUCTURING THE DEBATE: DELEGATION VS. THE SOVEREIGN MONOPOLY ON FORCE

- Letters of Marque (presently) = Authority + Capacity without Governance
- Gaps – Authority and Geopolitical Alignment (Strategic); Capacity, Targeting and Escalation Control (Operational); and Legal and Institutional Control (Governance)
- Framework grounding: Jensen & Meckling 1976 (principal-agent) and Avant 2005 (conditions for proxy force viability) | Thomson 1994 and Weber 1919 (state monopoly on force)

	Authority	Capacity	Governance
U.S. Government	✓ Art. I, Title 10, Title 50	X 50:1 against China alone	?
Private Sector	X CFAA says no	✓ Scale, tempo, attribution	?
Current Bills	✓ Art. I authorization	✓ Deputized private operators	X Not addressed

YO HO, YO HO, A BILL FOR ME

- H.R. 4988 (*Schweikert, Aug 2025*): Active defense carve-outs for foreign scam farms
- H.R. 9697 (*Burchett, Jul 2026*): Formal Letters of Marque for designated cyber threats
- S. 5000 (*Lee, Jul 2026*): Senate companion; establishes **15% bounty** on recovered foreign assets



COMPOUNDING HAZARDS OF DELEGATED FORCE

Constitutional Foundation |
Art. I, §8, Cl. 11 – Valid power,
but un-updated since 1856
(Declaration of Paris unratified)

Hazards:

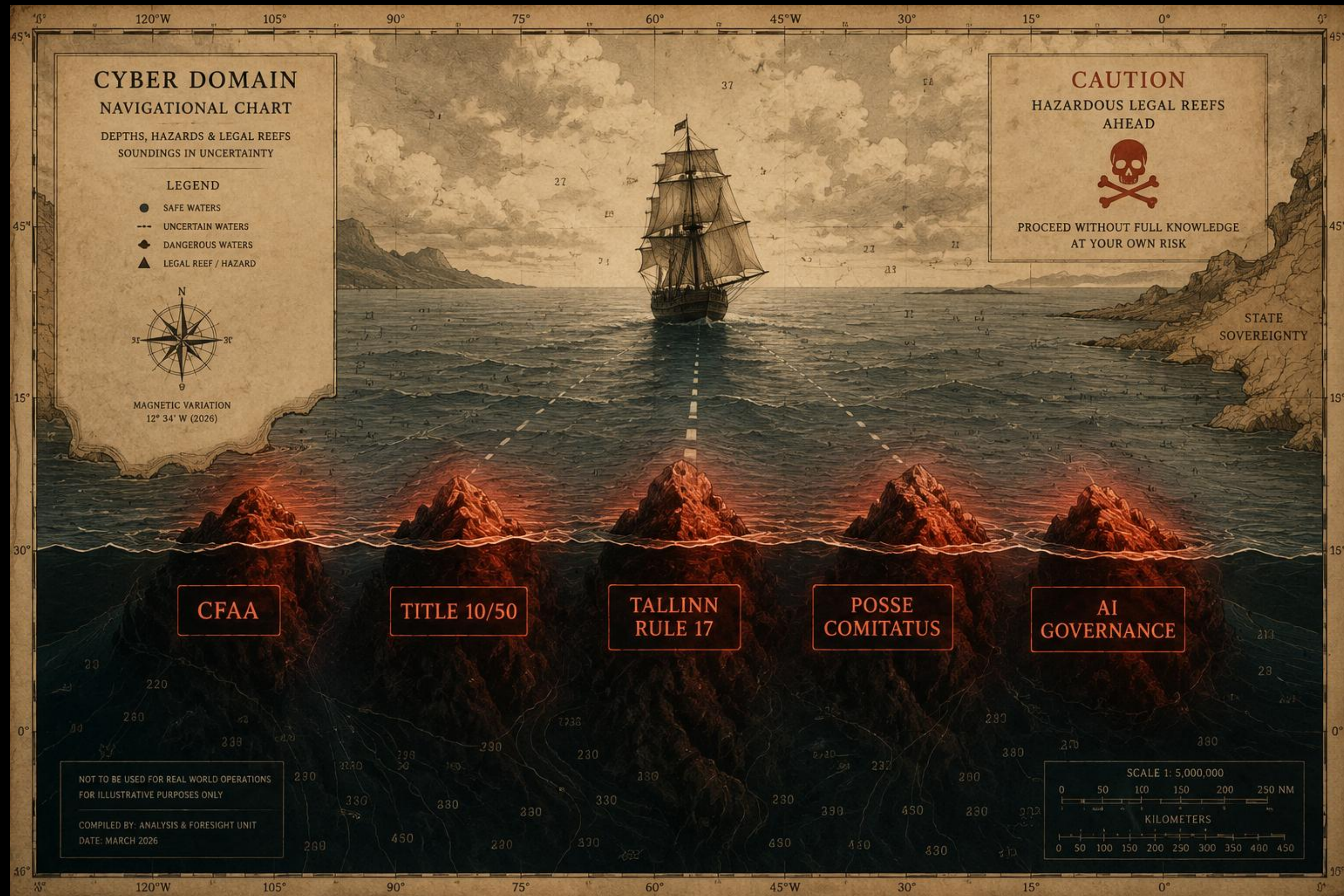
CFAA Lock | 18 U.S.C. § 1030
– “Outside U.S.” clause
creates jurisdictional ambiguity.

Title 10/50 Clash |
Deconfliction with classified
military/IC operations

**State Responsibility | Tallinn
Manual Rule 17** – Attaches
state liability where ops are
attributable under int’l law.

Domestic Military Limits |
*Posse Comitatus (18 U.S.C. §
1385)* – Limits domestic
military participation in law
enforcement

AI Governance – framework
ambiguity



WHY DID STATES TURN TO PIRATES?

- Sovereign capacity insufficient
- Logic: private actors already had the ships, the sailors, the knowledge of enemy waters, and the financial incentive to act – the state provided legal cover and shared the proceeds
- Letters of marque were not idealism; they were a **capacity patch**.



IN CONGRESS,

APRIL 3, 1776.

RESOLVED, That blank Commissions for private Ships of War, and Letters of Marque and Reprisal, signed by the President, be sent to the General Assemblies, Conventions, and Councils or Committees of Safety of the United Colonies, to be by them filled up and delivered to the Persons intending to fit out such private Ships of War for making Captures of British Vessels and Cargoes, who shall apply for the same, and execute the Bonds which shall be sent with the said Commissions, which Bonds shall be returned to the Congress.

By Order of Congress,

John Hancock PRESIDENT.

WHY EVERY NAVY EVENTUALLY FIRED ITS PIRATES

Why States Abandoned Privateering

Cyber Analogue

Professional navies built in-house

USCYBERCOM (U.S. Cyber Command) expansion — the capacity-building answer

Insurance markets refused coverage

Cyber insurers already pushing back on hack-back policies

Neutral shipping attacks damaged alliances

Five Eyes (U.S., UK, Canada, Australia, NZ intelligence alliance) infrastructure spillover — same failure mode

State legitimacy: mercenary profit undermined sovereign standing (*Thomson, 1994*)

Formal license publicly brands U.S. sovereignty as for hire

Industrial warfare exceeded private incentive structures

AI makes massive scale achievable — governance requirements scale with it



THE ADVERSARY MODEL: THEY FLY NO FLAG WHILE WE WOULD FILE A CONGRESSIONAL RECORD

**DENIABLE.
INVISIBLE.**

PIRATES DON'T FLY FLAGS.

ANONYMOUS VESSELS. UNTRACEABLE PURPOSE. PLAUSIBLE DENIAL.

**AUTHORIZED.
EXPOSED.**

A LICENSE IS A TARGET.

OFFICIAL LICENSE. TRACEABLE ENTITY. ZERO PLAUSIBLE DENIAL.

Democratic transparency and accountability – feature or bug in this context?

Adversary feature: state-directed or –tolerated boundary deliberately blurred

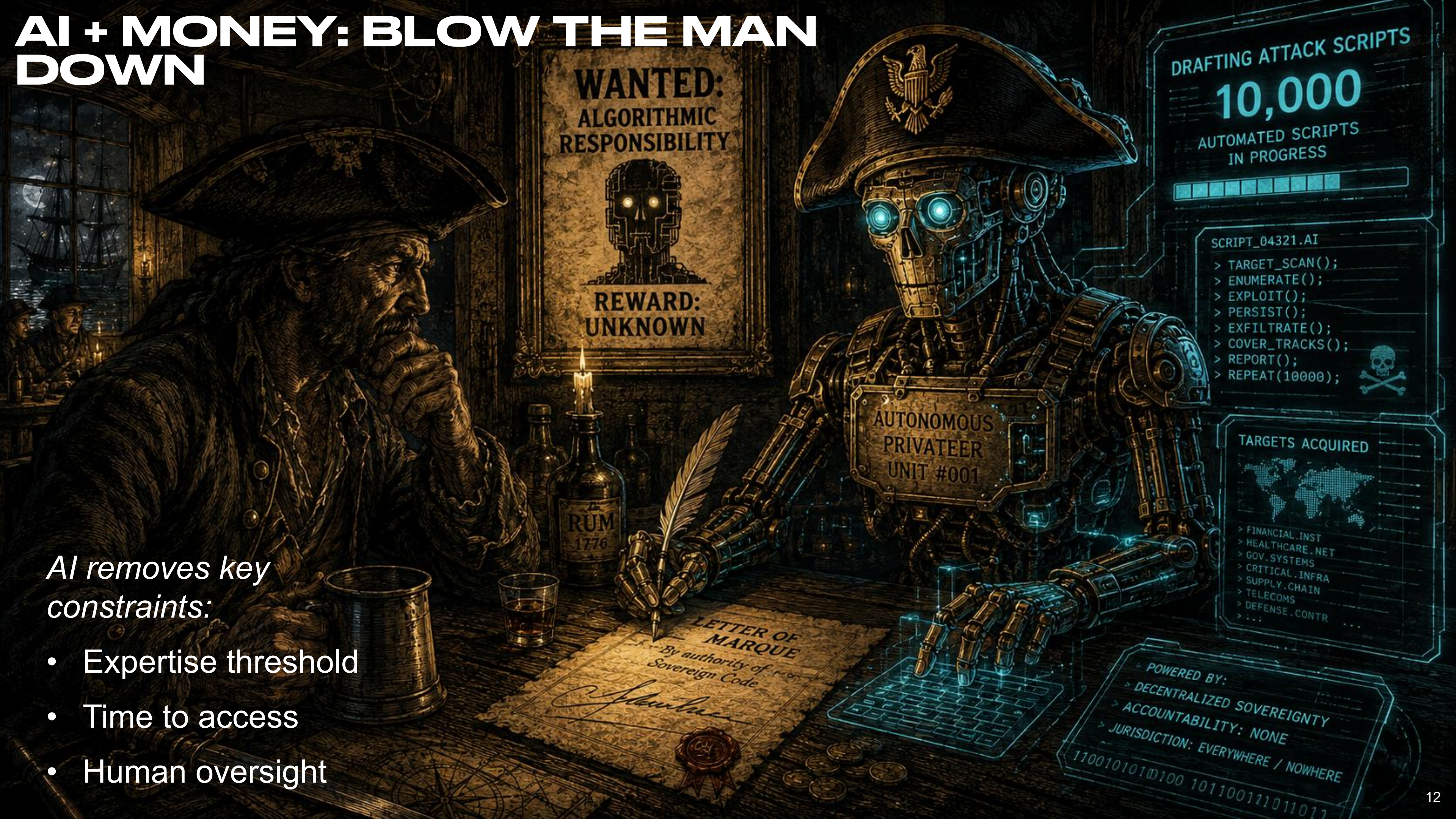
- PRC-linked – Haijun Minbing/People's Armed Forces Maritime Military; APT41; iSoon
- Iran – IRGC proxies and contractors
- Russia – RaaS and cyber TOC; Revil, Sandworm, FSB tolerance

THE IRONY OF AUTHORITY: WHAT YOU AUTHORIZE, YOU EXPOSE.

AI + MONEY: BLOW THE MAN DOWN

AI removes key constraints:

- Expertise threshold
- Time to access
- Human oversight



ALL HANDS ON DECK: THE ARGUMENT FOR LETTERS OF MARQUE



UNLEASH THE PRIVATEERS

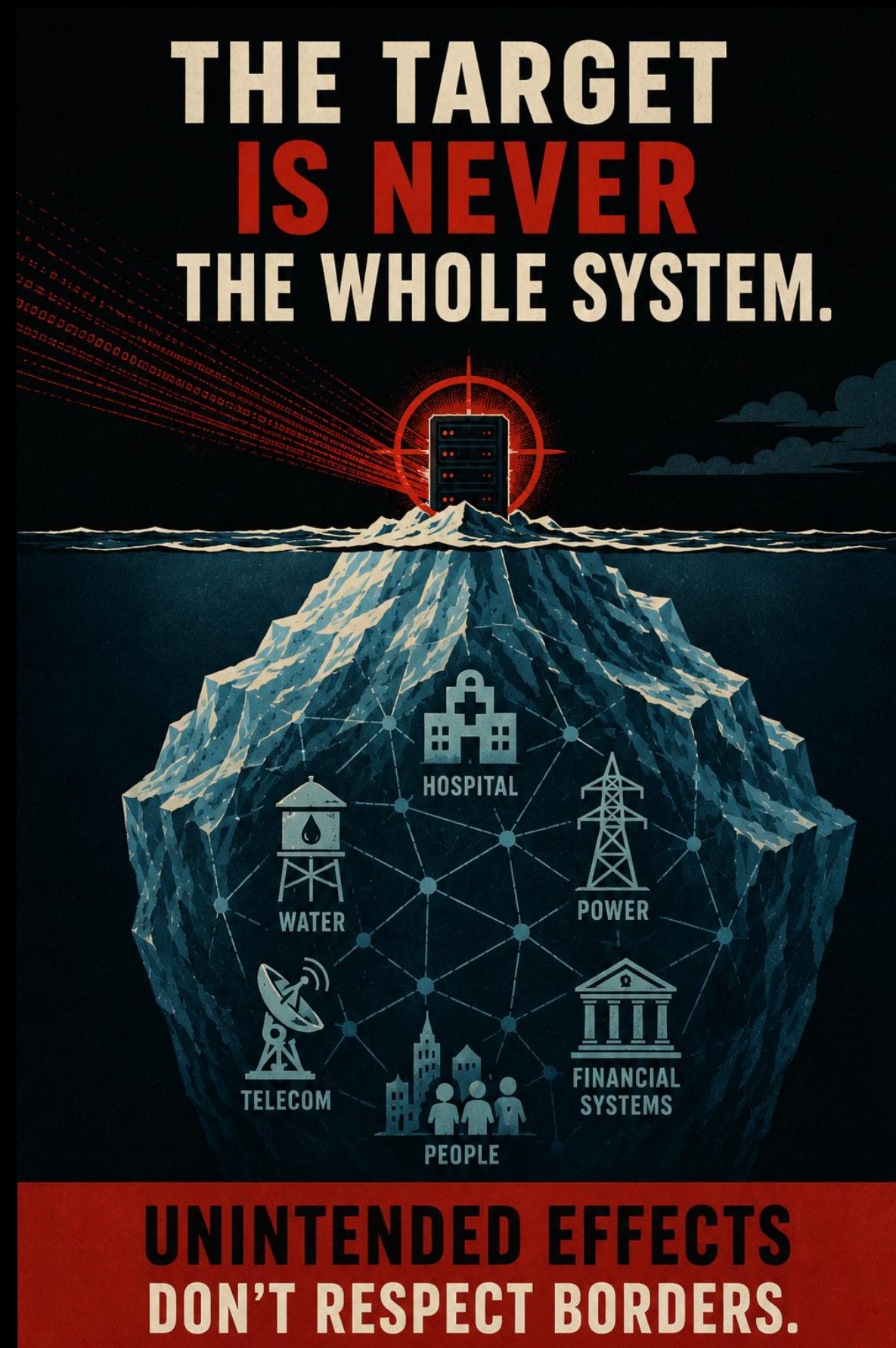


E S T . 2 0 4 9

- **Massive Scale Offset:**
Deputizes elite private capability against 50:1 adversary margins
- **Formalizing Gray Zones:** Can move existing/ongoing private disruptions into structured accountability; acknowledges reality of USG capabilities and America's need for defense
- **Imposing Economic Costs:**
Asset seizures force compellence where indictments and sanctions aren't achieving impact
- **Constitutional Precedent:**
Leverages explicit Article I authorities to adapt to the realities of asymmetric warfare

BATTEN DOWN THE HATCHES: THE ARGUMENT AGAINST LETTERS OF MARQUE

- ***We Have Alternatives:*** USG (CYBERCOM, NSA, IC, State, Treasury, DHS, Law enforcement), allies
- ***Shared Infrastructure:*** Cyber C2 is commingled – prize law ambiguity
- ***Equities Problem:*** Deconfliction likely very difficult
- ***Sovereignty/Alliance/Collateral Challenge:*** Associating USG with mercenary profit; signals on cyber norms; collateral damage
- ***Escalation Problem:*** Non-linear escalation through proxy compellence; NATO Art. 5 strategic ambiguity and potential triggers



FIVE STARS TO SAIL HER BY

Governance Factor	Pass If	Fail If	My Verdict
Control <i>Attribution survives delegation?</i>	Direct state command-and-control with pre-approved targeting.	Post-hoc reporting or unmonitored private targeting.	
Liability <i>Allies/Public tolerate collateral?</i>	Strict liability stack with full sovereign indemnity.	Private immunity shields that outsource legal risk.	
Scope <i>AI ops remain governable?</i>	Narrowly defined targets with strict rules of engagement.	Broad target categories with unchecked agentic execution.	
Incentives <i>Incentives stay aligned?</i>	Fixed service contracts with state-managed asset recovery.	Bounty splits or private retention of seized digital assets rather than return to victims	
De-confliction <i>Intel equities survive?</i>	Mandatory pre-strike interagency clearinghouse clearance.	Siloed, uncoordinated private offensive strikes.	

Reflected above are non-comprehensive but illustrative elements most relevant to ongoing legislative and policy debates. Framework of factors around governance evaluation derived from: principal-agent theory (Jensen & Meckling, 1976) | Tallinn Manual Rule 17 | Thomson 1994 | Avant 2005 | RAND / Brookings AI governance scholarship

THE HORIZON IS CLOSER THAN IT LOOKS – AND THE LAW STILL SAYS NO – SO WHERE TO NOW?

Privateers exist today, in gray zones.

The actual decision: The actual decision: formalize and regulate, accepting responsibility and building governance - or accept the costs of the status quo.

The Three Gaps test – does it solve Authority, Capacity, and Governance? (current bills address the first two only)

Democracies are unlikely to avoid the delegation question as cyber operations become increasingly automated and economically asymmetric.

THREE THINGS ANY ANSWER MUST DO:

Keep authority where accountability sits

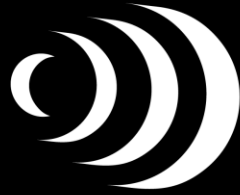
Structure what the private sector is already doing informally

Pass the five governance tests before you call it a solution



Thank you



 **PENUMBRA**
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REFERENCES

Legislation

[H.R. 9697/S. 5000 \(July 2026\) | H.R. 4988 \(August 2025\)](#)

Policy & Strategy

[Trump Cyber Strategy for America \(March 2026\) | WilmerHale Cyber Alert \(March 2026\) | Recorded Future News, Prague Conference \(March 19, 2026\) | Aspen Digital, So You Want to Hack Back \(October 2025\) | CSIS Back & Forth 4 \(April 2025\) | Lawfare, America Wants to Hack the Planet | Stewart Baker / Steptoe, The Hackback Debate](#)

Legal

[18 U.S.C. § 1030 \(CFAA\) | 18 U.S.C. § 1385 \(Posse Comitatus Act\) | Tallinn Manual 2.0 \(2017\), Rule 17 | Nicaragua v. U.S., ICJ \(1986\) | Tadic, ICTY \(1999\) | NDAA § 1642 \(2018\), defend forward authority | IU Law Review on Cyber Letters of Marque \(2026\) | NATO Article 5 and Cyber, CCDCOE | CISA 2015 \(Cybersecurity Information Sharing Act\)](#)

Threat Data

[FBI IC3 2025 Annual Report | Wray, Vanderbilt Summit on Modern Conflict \(April 2024\) | World Cybercrime Index, PLOS ONE \(2024\) | Salt Typhoon — Global Cyber Alliance Report \(December 2025\) | Volt Typhoon — CISA Advisory AA24-038A \(February 2024\) | Cybersecurity Ventures 2024 Global Cost Estimate | KK Park / Chainalysis Crypto Crime Report \(2024\)](#)

Scholarly

[Weber, Politics as a Vocation \(1919\) | Jensen & Meckling, Theory of the Firm \(1976\) | Olson, Logic of Collective Action \(1965\) | Schelling, Arms and Influence \(1966\) | Thomson, Mercenaries Pirates and Sovereigns \(1994\) | Avant, The Market for Force \(2005\) | Singer, Corporate Warriors \(2003\)](#)

Adversary Analysis

[CSIS AMTI Maritime Militia | Naval War College, Chinese Maritime Militia \(2020\) | Atlantic Council 5x5 China Cyber | Sekoia iSoon Ecosystem Report \(2024\)](#)